

MOHAMMADREZA GOLKAR

AI Researcher | Machine Learning Engineer | Prospective PhD Student (AI Engineering)
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PROFESSIONAL SUMMARY

M.Sc. graduate in Artificial Intelligence Engineering (GPA 19.91/20) with hands-on research experience in Explainable AI, Computer Vision, Large Language Models, and Agentic AI. Developed interpretable deep learning pipelines for medical image analysis using EfficientNet, Vision Transformers, Grad-CAM, and SHAP, with fairness evaluation across skin-type subgroups, and built FairMedBench, an open-source benchmark for fairness, explainability, and calibration of medical image classifiers. Built AI agent and RAG systems with LangChain, LangGraph, and FAISS, including ResearchQA, an evaluation-centered RAG framework with claim-level citation verification. Over two years of industry software engineering experience. Seeking a PhD position in trustworthy, explainable, human-centered AI.

RESEARCH INTERESTS

Core: Explainable AI (XAI) • Trustworthy AI • Fairness & Bias Analysis • Medical Image Analysis

Also: Large Language Models • Foundation Models • Agentic AI & Multi-Agent Systems • Retrieval-Augmented Generation (RAG) • Computer Vision • Human-Centered AI

EDUCATION

M.Sc. in Artificial Intelligence Engineering 2024 – 2026

Islamic Azad University, Khomeinishahr – Isfahan, Iran

- GPA: 19.91/20 | Thesis grade: 18.5/20
- Thesis: Fair and Explainable Deep Learning for Skin Lesion Classification

B.Sc. in Information Technology Engineering 2012 – 2017

Islamic Azad University, Najafabad – Isfahan, Iran

RESEARCH EXPERIENCE

Graduate Researcher – Thesis Project 2024 – 2026

Islamic Azad University, Khomeinishahr – Isfahan, Iran | *Fair and Explainable Deep Learning for Skin Lesion Classification*

- Fine-tuned EfficientNetB0 and Vision Transformer (ViT) models on more than 10,000 dermatoscopic images across 7 skin lesion categories.
- Applied Grad-CAM and SHAP to produce saliency maps and feature importance explanations for clinical decision support.
- Designed and ran a fairness audit across 6 Fitzpatrick skin-type groups, quantifying and mitigating performance disparities in minority classes.
- Improved minority-class F1-score with mixup, cutout, and class-weighted sampling.
- Built reproducible PyTorch and TensorFlow pipelines covering preprocessing, training, evaluation, and explainability. Metrics: Accuracy, Precision, Recall, F1, AUC.

PUBLICATIONS

- Golkar, M. "Enhancing Clinical Trust in Deep Learning Models Through Grad-CAM-Based Explainability in Radiology Image Classification: A Critical Review of the Evidence and a Layered Evaluation Framework." *Manuscript under submission*, 2026.

PROJECTS

FairMedBench: Benchmark Framework for Fair, Explainable and Reliable Medical Image Classification 2026

Python, PyTorch, Grad-CAM, Integrated Gradients, YAML configs, Docker, pytest | github.com/FlowerMr/FairMedBench

- **Situation & Task:** Medical image classifiers are usually judged by overall accuracy alone, although they can perform differently across skin-tone groups, give unstable explanations, be poorly calibrated, or fail under small input changes. Build an open, reproducible research framework that evaluates these properties together.
- **Action:** Designed a modular Python package (data, models, metrics, fairness, explainability, uncertainty, robustness, reporting) with YAML-configured experiments and automated CSV/JSON/figure reports. Implemented subgroup fairness metrics (Equal Opportunity and Demographic Parity gaps), XAI (Grad-CAM, Integrated Gradients) with faithfulness and stability evaluation, calibration (temperature scaling, MC Dropout, ECE, Brier score), and perturbation-based robustness tests. Added a synthetic-data demo, dataset validation, unit tests, and Docker support.
- **Result:** Benchmarked ResNet18 and ViT-Tiny on the DDI dataset with Fitzpatrick skin-type groups. ViT-Tiny had higher accuracy (0.750 vs 0.674) and specificity (0.888 vs 0.755), but a larger Equal Opportunity gap (0.402 vs 0.311) and lower XAI faithfulness (0.018 vs 0.080), showing that higher accuracy did not mean fairer or more explainable behavior. Released open source (MIT).

ResearchQA: Citation-Verified RAG for Scientific Papers

2026

Python, FAISS, BM25, Sentence Transformers, Cross-Encoder, NLI (DeBERTa-v3), Ollama, Streamlit, Docker, pytest

- **Situation & Task:** LLM answers over scientific papers sound confident even when unsupported, and typical “chat with PDF” systems give no guarantee that a sentence actually comes from the cited passage. Build an open, reproducible RAG framework that links every answer sentence to its evidence, verifies the link, and evaluates retrieval, generation, and citation quality separately.
- **Action:** Implemented a full pipeline: structure-aware chunking with paragraph-level provenance, hybrid retrieval (dense FAISS + a custom BM25 with Reciprocal Rank Fusion), cross-encoder reranking, and local generation with Ollama. Answers are split into claims mapped to citations; each claim is checked with an NLI model plus lexical-overlap and numeric-consistency checks (supported / partially supported / unsupported). Unsupported claims are dropped or flagged, and the system abstains when evidence is insufficient. Built an evaluation harness on a 30-paper Qasper subset with Recall@K, MRR, nDCG, evidence coverage, citation support, abstention rate, bootstrap 95% confidence intervals, ablations (reranking, verification), and rule-based error analysis. Added a CLI, Streamlit interface, YAML-configured experiments, extensive pytest suite, and Docker support.
- **Result:** Delivered an open-source, modular framework (MIT) whose evaluation pipeline compares six retrieval and four generation configurations with ablations, and that reports its own limitations explicitly, including verifier error and evaluator circularity.

ResearchMatch-AI: Multi-Agent Research Position Matching System

2025

Python, LangGraph, LangChain, Sentence Transformers, FAISS, Streamlit

- **Situation & Task:** Finding PhD positions that fit a researcher's profile, and understanding what is missing for each one, is slow and manual. Build a multi-agent system that matches researchers with PhD positions and research opportunities.
- **Action:** Built 5 specialized agents with LangGraph and LangChain for CV analysis, skill extraction, research topic extraction, position analysis, and competency gap assessment. Applied semantic similarity with Sentence Transformers and FAISS retrieval over 1,000+ research position descriptions.
- **Result:** The system generates a personalized gap analysis report and application strategy recommendation for each candidate–position pair.

Local Secure RAG Chatbot for Organizational Knowledge

2025

Python, FAISS, Hugging Face Transformers, Ollama, LangChain

- **Situation & Task:** Organizations need to query private documents without sending data to cloud services. Build a fully local, privacy-preserving Retrieval-Augmented Generation system.
- **Action:** Implemented the full pipeline: document ingestion (PDF, DOCX, plain text), chunking, embedding with Sentence Transformers, FAISS vector storage, hybrid dense/sparse retrieval, and answer generation with Ollama. Added source-grounded generation and confidence-based retrieval filtering.
- **Result:** Reduced the hallucination rate through source-grounded answers and retrieval filtering, with no cloud data exposure.

Semantic Search Chatbot

2025

Python, FAISS, Sentence Transformers, Gradio

- **Situation & Task:** Keyword search (BM25) misses relevant text when the query wording differs from the document wording. Build a semantic retrieval system over a large text collection.
- **Action:** Developed dense vector search (Sentence Transformers + FAISS) over more than 10,000 text chunks and built an interactive Gradio interface for real-time queries and ranked document retrieval.
- **Result:** The semantic system outperformed BM25 keyword-based baselines and gave users a working real-time search interface.

Multilingual PDF Summarizer

2024

PyTorch, Hugging Face, NLLB-200, NLP

- **Situation & Task:** Long documents are slow to review for readers working across Persian, German, and English. Build a local pipeline that summarizes long documents and translates the summaries.
- **Action:** Developed a local LLM pipeline with automated document parsing, hierarchical chunking, abstractive summarization, and translation between Persian, German, and English using NLLB-200 models.
- **Result:** Delivered a fully automated, locally running workflow from a long PDF to a summary in the chosen language.

LLM Interview Summarization Analysis

2024

Hugging Face Transformers, Prompt Engineering, NLP

- **Situation & Task:** Summarizing real interview transcripts is hard, and it is unclear how prompting strategy affects summary quality. Evaluate how FLAN-T5 and BART summarize real-world interview transcripts under different prompting conditions.
- **Action:** Compared zero-shot and few-shot prompting on both models and analyzed factual consistency, coherence, and abstraction quality.
- **Result:** Produced a comparative analysis of the two models and the two prompting strategies across these three quality dimensions.

INDUSTRY EXPERIENCE

Software Engineer – Mobile Applications, Megashid (Iran)

Sep 2023 – Sep 2024

- Developed and shipped Android applications in Kotlin and Flutter, serving thousands of active users.
- Worked on Industry 4.0 and IoT product integrations with cross-functional teams; refactored memory-intensive modules, reducing crash rate and improving responsiveness.

Software Engineer – Mobile Applications, Terapico (Iran)

Mar 2022 – Aug 2023

- Developed Android applications for digital healthcare platforms serving end users in Iran.
- Implemented scalable features using MVVM, REST APIs, and Firebase; collaborated with product and engineering teams across design, development, QA, and release.

MILITARY SERVICE

Iranian Air Force (Army) – Mandatory national service

Feb 2019 – Nov 2020

TECHNICAL SKILLS

Dev & Tools: Python, SQL, Kotlin, Bash, Git, Linux, Docker (basic), Gradio, Streamlit, Jupyter, W&B, pytest, REST APIs, Firebase, n8n

ML & Computer Vision: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, EfficientNet, ResNet, Vision Transformers (ViT)

LLMs, Agents & RAG: LangChain, LangGraph, Ollama, OpenAI API, Prompt Engineering, FAISS, Sentence Transformers, Vector Databases, BM25, Hybrid Retrieval (RRF), Cross-Encoder Reranking, NLI-based Claim Verification, RAG Evaluation

Explainable AI & Fairness: Grad-CAM, SHAP, Integrated Gradients, Fairness & Bias Analysis, Calibration (ECE, Temperature Scaling)

Languages: Persian (native) | English (TOEFL iBT 90/120, CEFR B2/C1) | German (B2)